AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

MT395
Applied Cybersecurity

Final Examination
Summer Semester 2025/2026

Time Allowed: 2 Hours

AOU EGYPT

Instructions:

- Total Marks: **50**
- This exam consists of **5** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are **Not** allowed.

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part 1: Choose The correct answer [10 marks]

1. A disgruntled software developer embeds code inside a company's production application that deletes customer archives if the developer's employee ID is removed from the active payroll database. This is an example of a:
   A) Worm
   B) Spyware
   C) Logic Bomb
   D) Rootkit
2. During a Denial of Service (DoS) attack, an external host floods a target web server with a massive volume of TCP connection initializations but purposefully ignores the server's SYN-ACK responses. This specific attack is a:
   A) MAC Flooding Attack
   B) SYN Flood
   C) Buffer Overflow
   D) Smurf Attack
3. A company enforces a system where an employee's computer must automatically have its antivirus signatures checked and OS patches validated before it is allowed onto the main corporate subnet. This architecture is called:
   A) Data Loss Prevention (DLP)
   B) Network Access Control (NAC)
   C) Unified Threat Management (UTM)
   D) Stateless Filtering
4. Which type of virtualization layer installs and executes directly on a computer's physical bare-metal hardware infrastructure without requiring an underlying host operating system?
   A) Type 2 Hypervisor
   B) Type 1 Hypervisor
   C) Application Container
   D) Hosted Sandbox
5. To achieve robust wireless local network confidentiality and data integrity, the WPA2 standard mandates the implementation of which mathematical block cipher algorithm?
   A) RC4
   B) 3DES
   C) AES (Advanced Encryption Standard)
   D) MD5
6. An application development team uses automated testing software to feed arbitrary, malformed, and randomized data blocks into an application's input fields to check for hidden memory leaks or structural crashes. This process is called:
   A) Static Code Analysis
   B) Fuzzing (Fuzz Testing)
   C) Application Whitelisting
   D) Input Code Signing
7. An enterprise allows its staff members to use their personally owned tablets and mobile phones to sync corporate emails and view internal documentation. This deployment model is called:
   A) COPE
   B) BYOD (Bring Your Own Device)
   C) CYOD
   D) VDI
8. An enterprise environment implements a solution that enables users to input a single set of network credentials one time to safely navigate across multiple independent corporate software platforms. This mechanism is called:
   A) Multi-Factor Authentication (MFA)
   B) Single Sign-On (SSO)
   C) Cognitive Biometrics

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D) Federated Group Policy

9. Which security access control model operates on the rule that the explicit creator or owner of a data resource retains absolute discretion to grant or revoke system rights to other network users?
   A) Mandatory Access Control (MAC)
   B) Discretionary Access Control (DAC)
   C) Role-Based Access Control (RBAC)
   D) Attribute-Based Access Control (ABAC)
10. Which physical sanitation technique uses a specialized device to disrupt or neutralize the internal magnetic field of a faulty hard disk drive, making data retrieval permanently impossible?
   A) Wiping
   B) Shredding
   C) Degaussing
   D) Overwriting

Part 2: Case Studies

Case Study (1) **(4 Mark)**

A medical data processing company suffers an internal breach when a front-desk operator inserts a personal USB drive into an office client machine to run an unauthorized, third-party software utility (Sideloading). The utility carries an embedded piece of spyware that harvests host credentials and uses them to modify confidential master databases. An investigation reveals that the operator's account possessed full administrative write access across all master servers, despite their daily duties being limited to entering client name records. Furthermore, no system was in place to audit endpoint application compliance or catch sensitive files leaving the system.

1. Which fundamental identity and access management principle was violated here, and how should it be corrected? (2 Mark)
2. Name two endpoint or data-centric technologies that could have stopped this incident.(2 Mark)

Case Study (2) **(6 Mark)**

A security team discovers that a wireless system located at a remote operations facility was easily compromised by an external driver parked near the exterior fence line. The attacker leveraged automated cracking tools to iterate through sequential numerical authentication protocols exposed by the router's enabled Wi-Fi Protected Setup (WPS) PIN engine. Once inside the wireless subnet, the attacker used an unauthorized laptop to poison the gateway address maps, positioning their device to systematically intercept unencrypted traffic from internal industrial sensor devices.

1. Identify the entry vulnerability exploited by the attacker. (2 Mark)
2. Name the secondary attack mechanism executed once the attacker gained subnet access. (2 Mark)
3. Propose an immediate remediation strategy. (2 Mark)

Part 3: Answer the following questions

1. List the five primary authentication credential categories used to validate an individual's identity. (2 Mark)
2. Compare Mobile Device Management (MDM) and Mobile Application Management (MAM) regarding their architectural focus and corporate deployment suitability. (4 Mark)

3. Compare a Vulnerability Scan with a Penetration Test based on operational methodology and core objectives. (4 Mark)

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Part 4: Answer the following questions

1. Define "Accounting" as it relates to Identity and Access Management terminology. (2 Mark)

2. List four essential tools or processes used during secure hardware system initialization to establish client endpoint security. (2 Mark)

3. Explain the operational mechanics of a "SYN Flood" attack and how it achieves a Denial of Service (DoS). (3 Mark)

4. Explain why data should never be sanitized using the standard operating system "delete" command, and explain how "Degaussing" solves this issue for magnetic drives. (3 Mark)

Part 4, Match the cybersecurity term with its corresponding definition: [10 Marks]

No.	Security Term	Definition / Description
1	**Obscurity**	**A.** A dedicated hardware microchip embedded on a motherboard to securely generate and store cryptographic keys.
2	**Rootkit**	**B.** An access control model where permissions are granted dynamically based on automated parameters like user location or department role.
3	**SYN Flood**	**C.** A wireless protocol that delivers robust local network privacy by mandating Advanced Encryption Standard (AES) block encryption.
4	**VLAN**	**D.** A specialized security infrastructure that audits device health baselines and patch status before permitting full network connection.
5	**Type 1 Hypervisor**	**E.** A data destruction process that completely neutralizes magnetic fields to permanently wipe hard disk drives.
6	**WPA2**	**F.** A server-focused attack that exhausts system connection queues by dropping half-open TCP handshakes.
7	**TPM**	**G.** A strategy that relies purely on hiding internal configurations, versions, or system architecture to impede an adversary.
8	**SCADA**	**H.** Virtualization software that installs and executes directly on raw bare-metal hardware without a host operating system.
9	**Network Access Control / NAC**	**I.** Industrial data collection and automated control architectures used to monitor critical utilities like electrical grids.
10	**Degaussing**	**J.** Advanced malware designed to hide active processes and malicious modules by directly subverting lower kernel layers of the OS.
		K. A technology that segments a physical switch infrastructure into logically separated, software-defined broadcast domains.

End of questions

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